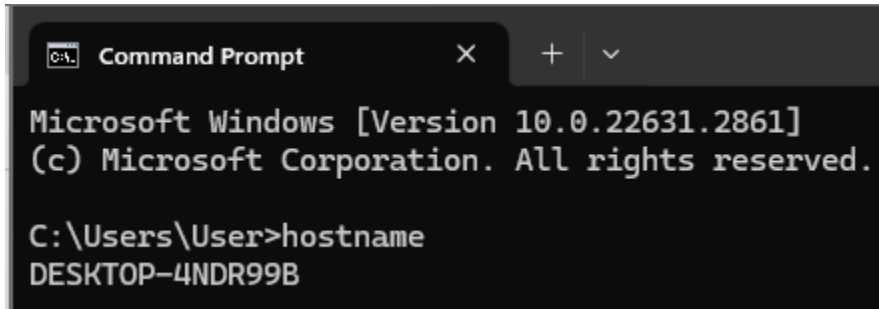


Windows Hardening by Niyaz Mursid Dolon

Part A: Windows Command Line Administration — Practical Exercise.

1. Find and display the system hostname .

hostname

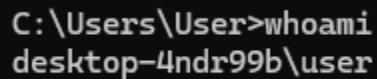
A screenshot of a Windows Command Prompt window. The title bar says "Command Prompt". The text inside shows the Windows version and copyright information, followed by the command "hostname" and its output "DESKTOP-4NDR99B".

```
Microsoft Windows [Version 10.0.22631.2861]
(c) Microsoft Corporation. All rights reserved.

C:\Users\User>hostname
DESKTOP-4NDR99B
```

2. Show the current logged-in username.

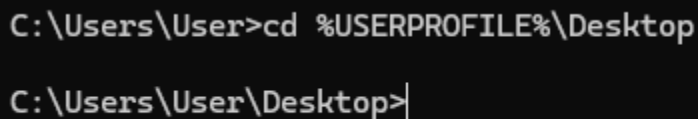
Whoami

A screenshot of a Windows Command Prompt window showing the command "whoami" and its output "desktop-4ndr99b\user".

```
C:\Users\User>whoami
desktop-4ndr99b\user
```

3. Navigate to the Desktop directory using Windows Command Line (CMD).

cd %USERPROFILE%\Desktop

A screenshot of a Windows Command Prompt window showing the command "cd %USERPROFILE%\Desktop" and the resulting directory path "C:\Users\User\Desktop".

```
C:\Users\User>cd %USERPROFILE%\Desktop

C:\Users\User\Desktop>
```

4. Create a new folder named Practice1 and Practice2 on the Desktop.

mkdir Practice1 Practice2

```
C:\Users\User\Desktop>mkdir Practice1 Practice2  
  
C:\Users\User\Desktop>
```

5. **Create a text file named info1.txt and info2.txt inside Practice1.**
cd Practice1
echo > info1.txt
echo > info2.txt

```
C:\Users\User\Desktop>cd Practice1  
  
C:\Users\User\Desktop\Practice1> echo > info1.txt  
  
C:\Users\User\Desktop\Practice1> echo > info2.txt
```

6. **List all files and folders inside Practice1.**
dir

```
C:\Users\User\Desktop\Practice1>dir  
Volume in drive C is Win-11  
Volume Serial Number is A0DF-8980  
  
Directory of C:\Users\User\Desktop\Practice1  
  
11/28/2025  12:10 AM    <DIR>          .  
11/27/2025  11:57 PM    <DIR>          ..  
11/28/2025  12:09 AM                13 info1.txt  
11/28/2025  12:10 AM                13 info2.txt  
                2 File(s)                26 bytes  
                2 Dir(s)  54,757,937,152 bytes free
```

7. **Write the sentence “Windows CMD Assignment” inside info1.txt.**
echo Windows CMD Assignment > info1.txt

```
C:\Users\User\Desktop\Practice1>echo Windows CMD Assignment > info1  
.txt  
  
C:\Users\User\Desktop\Practice1>
```

8. **Copy info1.txt to the under folder Practice2.**

copy info1.txt ..\Practice2

```
C:\Users\User\Desktop\Practice1>copy info1.txt ..\Practice2
1 file(s) copied.
```

9. **Rename info2.txt to final.txt.**

ren info2.txt final.txt

```
C:\Users\User\Desktop\Practice1>ren info2.txt final.txt

C:\Users\User\Desktop\Practice1>
```

10. **Move the final.txt from folder Practice2 to folder Practice1.**

Cd Practice2

ren info1.txt final.txt

move final.txt ..\Practice1

```
C:\Users\User\Desktop>Cd Practice2

C:\Users\User\Desktop\Practice2>move final.txt ..\Practice1
The system cannot find the file specified.

C:\Users\User\Desktop\Practice2>Ren info1.txt final.txt

C:\Users\User\Desktop\Practice2>Move final.txt ..\Practice1
Overwrite C:\Users\User\Desktop\Practice1\final.txt? (Yes/No/All):yes
1 file(s) moved.
```

11. **Delete the file final.txt.**

del final.txt

```
C:\Users\User\Desktop\Practice2>cd..  
  
C:\Users\User\Desktop>cd Practice1  
  
C:\Users\User\Desktop\Practice1>del final.txt  
  
C:\Users\User\Desktop\Practice1>|
```

12. Check the current system date and time.

date /t
time /t

```
C:\Users\User\Desktop\Practice1>date /t  
Fri 11/28/2025  
  
C:\Users\User\Desktop\Practice1>time /t  
01:05 AM
```

13. Change the system directory to C:\Windows.

cd C:\Windows

```
C:\Users\User\Desktop\Practice1>cd C:\Windows  
  
C:\Windows>|
```

14. List all hidden files in the C:\Windows directory.

dir /a:h

```
C:\Windows>dir /a:h  
Volume in drive C is Win-11  
Volume Serial Number is A0DF-8980  
  
Directory of C:\Windows  
  
11/23/2025  01:13 PM    <DIR>          ..  
05/07/2022  04:16 PM    <DIR>          BitLockerDiscoveryVolumeContents  
05/07/2022  11:24 AM    <DIR>          ELAMBKUP  
11/23/2025  01:14 PM    <DIR>          Installer  
05/07/2022  11:24 AM    <DIR>          LanguageOverlayCache  
05/07/2022  11:19 AM                670 WindowsShell.Manifest  
               1 File(s)                670 bytes  
               5 Dir(s)  54,738,669,568 bytes free
```

15. Display your system's IP configuration details.

`ipconfig /all`

```

C:\Windows>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : DESKTOP-4NDR99B
    Primary Dns Suffix . . . . . :
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No


Ethernet adapter Ethernet 5:

    Connection-specific DNS Suffix . :
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-00-27-00-00-04
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::e2c3:fbaf:e52b:1087%4(Preferred)
    IPv4 Address. . . . . : 192.168.56.1(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
    DHCPv6 IAID . . . . . : 537526311
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-52-56-3A-18-C0-4D-9D-57-06
    NetBIOS over Tcpip. . . . . : Enabled


Ethernet adapter Ethernet 4:

    Connection-specific DNS Suffix . :
    Description . . . . . : Realtek Gaming 2.5GbE Family Controller #2
    Physical Address. . . . . : CC-28-AA-07-53-4F
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::65f6:11cf:22d6:31f5%13(Preferred)
    IPv4 Address. . . . . : 192.168.0.106(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : Thursday, November 27, 2025 7:56:06 PM
    Lease Expires . . . . . : Friday, November 28, 2025 2:46:38 AM
    Default Gateway . . . . . : 192.168.0.1
    DHCP Server . . . . . : 192.168.0.1
    DHCPv6 IAID . . . . . : 365701290
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-52-56-3A-18-C0-4D-9D-57-06
    DNS Servers . . . . . : 192.168.0.1
    NetBIOS over Tcpip. . . . . : Enabled

```

16. Ping www.google.com to check internet connectivity.

ping www.google.com

```
C:\Windows>ping www.google.com

Pinging www.google.com [172.217.24.4] with 32 bytes of data:
Reply from 172.217.24.4: bytes=32 time=28ms TTL=113
Reply from 172.217.24.4: bytes=32 time=30ms TTL=113
Reply from 172.217.24.4: bytes=32 time=30ms TTL=113
Reply from 172.217.24.4: bytes=32 time=30ms TTL=113

Ping statistics for 172.217.24.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 30ms, Average = 29ms
```

17. Check the list of all running processes using CMD.

Tasklist

```
C:\Windows>Tasklist
```

Image Name	PID	Session Name	Session#	Mem Us
System Idle Process	0	Services	0	
8 K				
System	4	Services	0	14
8 K				
Secure System	172	Services	0	46,68
4 K				
Registry	244	Services	0	31,65
6 K				
smss.exe	736	Services	0	1,13
2 K				
csrss.exe	1064	Services	0	5,24
4 K				
wininit.exe	1160	Services	0	6,35
2 K				
services.exe	1232	Services	0	9,28
0 K				
LsaIso.exe	1252	Services	0	3,57
2 K				
lsass.exe	1268	Services	0	28,34
4 K				
svchost.exe	1448	Services	0	30,10
0 K				
fontdrvhost.exe	1476	Services	0	3,18
8 K				
svchost.exe	1580	Services	0	16,80
4 K				

18. Display a tree structure of directories under C:\.

tree C:\

```

2506_none_04e316ab08507b18
|   |   |   |   |
|   |   |   |   |---f
|   |   |   |   |---r
|   |   |   |   |---wow64_microsoft-windows-cmi_31bf3856ad364e35_10.0.22621.1_non
e_15b8c04e82a9d0d3
|   |   |   |   |---wow64_microsoft-windows-cngcredui-dll_31bf3856ad364e35_10.0.2
2621.1_none_bebfbfe4425334c1
|   |   |   |   |---wow64_microsoft-windows-com-base-qfe-ole32_31bf3856ad364e35_1
0.0.22621.2506_none_964cb745a251aa80
|   |   |   |   |---f
|   |   |   |   |---r
|   |   |   |   |---wow64_microsoft-windows-com-base_31bf3856ad364e35_10.0.22621.
2792_none_f8110442d68d7dac
|   |   |   |   |---f
|   |   |   |   |---r
|   |   |   |   |---wow64_microsoft-windows-com-coml2_31bf3856ad364e35_10.0.22621
.2506_none_07ec63ea2df7f297
|   |   |   |   |---f
|   |   |   |   |---r
|   |   |   |   |---wow64_microsoft-windows-com-complus-admin_31bf3856ad364e35_10
.0.22621.1_none_49542a9554ec25db
|   |   |   |   |---wow64_microsoft-windows-com-complus-msc_31bf3856ad364e35_10.0
.22621.1_none_91876d8812de4f17
|   |   |   |   |---wow64_microsoft-windows-com-complus-runtime_31bf3856ad364e35_
10.0.22621.1_none_c5b53517fcd07fc4
|   |   |   |   |---wow64_microsoft-windows-com-complus-setup_31bf3856ad364e35_10
.0.22621.1_none_55924f4f176d2089

```

19. View system information on a Windows operating system.

Systeminfo

```

Host Name:                DESKTOP-4NDR99B
OS Name:                  Microsoft Windows 11 Pro
OS Version:               10.0.22631 N/A Build 22631
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         User
Registered Organization:
Product ID:                00331-10000-00001-AA239
Original Install Date:     8/17/2024, 6:48:52 PM
System Boot Time:          11/27/2025, 2:23:06 PM
System Manufacturer:       ASUS
System Model:              System Product Name
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 183 Stepping 1 Ge
nuineIntel ~2500 Mhz

```


20. **View the version of your Windows operating system.**

Ver

```
C:\Windows>Ver
```

```
Microsoft Windows [Version 10.0.22631.2861]
```